

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

19
20

COURSE NAME:CSA0702 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

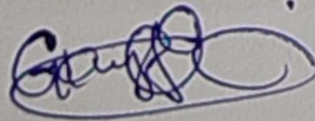
CONCEPT MAPPING

Code of Conduct:

I G.K.VYSHNAVI/192425069 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Q1. Missing Layers in the OSI Model

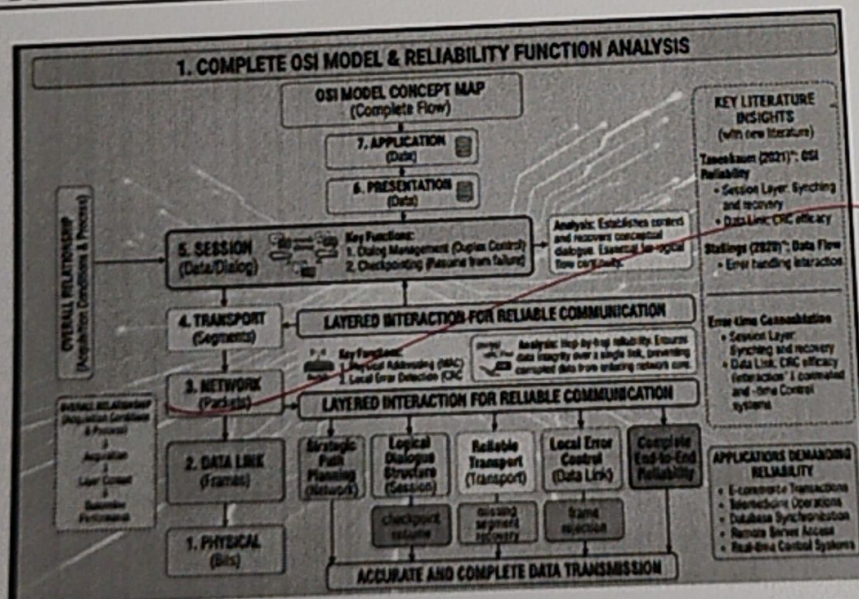
The missing layers are:

- Session Layer
- Data Link Layer

The OSI (Open Systems Interconnection) model consists of seven layers that enable reliable communication between devices. The Session Layer establishes, manages, and terminates communication sessions between applications. It ensures synchronization and maintains communication checkpoints during long data transfers. The Data Link Layer is responsible for node-to-node communication, framing, MAC addressing, error detection, and flow control. Without these two layers, communication cannot be properly synchronized or delivered reliably over the physical medium.

Concept Map Architecture

Source Concept	Relationship	Target Concept
Application	passes data to	Presentation
Presentation	passes data to	Session Layer
Session Layer	connects with	Transport Layer
Transport Layer	passes data to	Network Layer
Network Layer	passes data to	Data Link Layer
Data Link Layer	transmits through	Physical Layer



Literature Insights & Evidence (Q1)

Kurose & Ross (2021): Describe the conceptual differences between the theoretical OSI reference model and the practical Internet (TCP/IP) architecture.

Tanenbaum & Wetherall (2019): Explain how Layer 5 (Session) establishes dialogue control and handles checkpoint recovery, noting how these functions are absorbed by application-level protocols today.

Stallings (2016): Discusses node-to-node reliability, detailing how Layer 2 (Data Link) utilizes framing, flow control, and error detection over a single physical link.

Forouzan (2021): Covers the complete data encapsulation processing chain as bits are transformed from logical application sessions down to physical media streams.

Q2. Data Units and OSI Layers

The OSI model uses different Protocol Data Units (PDUs) while transmitting information. During encapsulation, data is transformed into Segments, Packets, Frames, and Bits as it moves through the layers. Each transformation adds protocol-specific information required for successful communication.

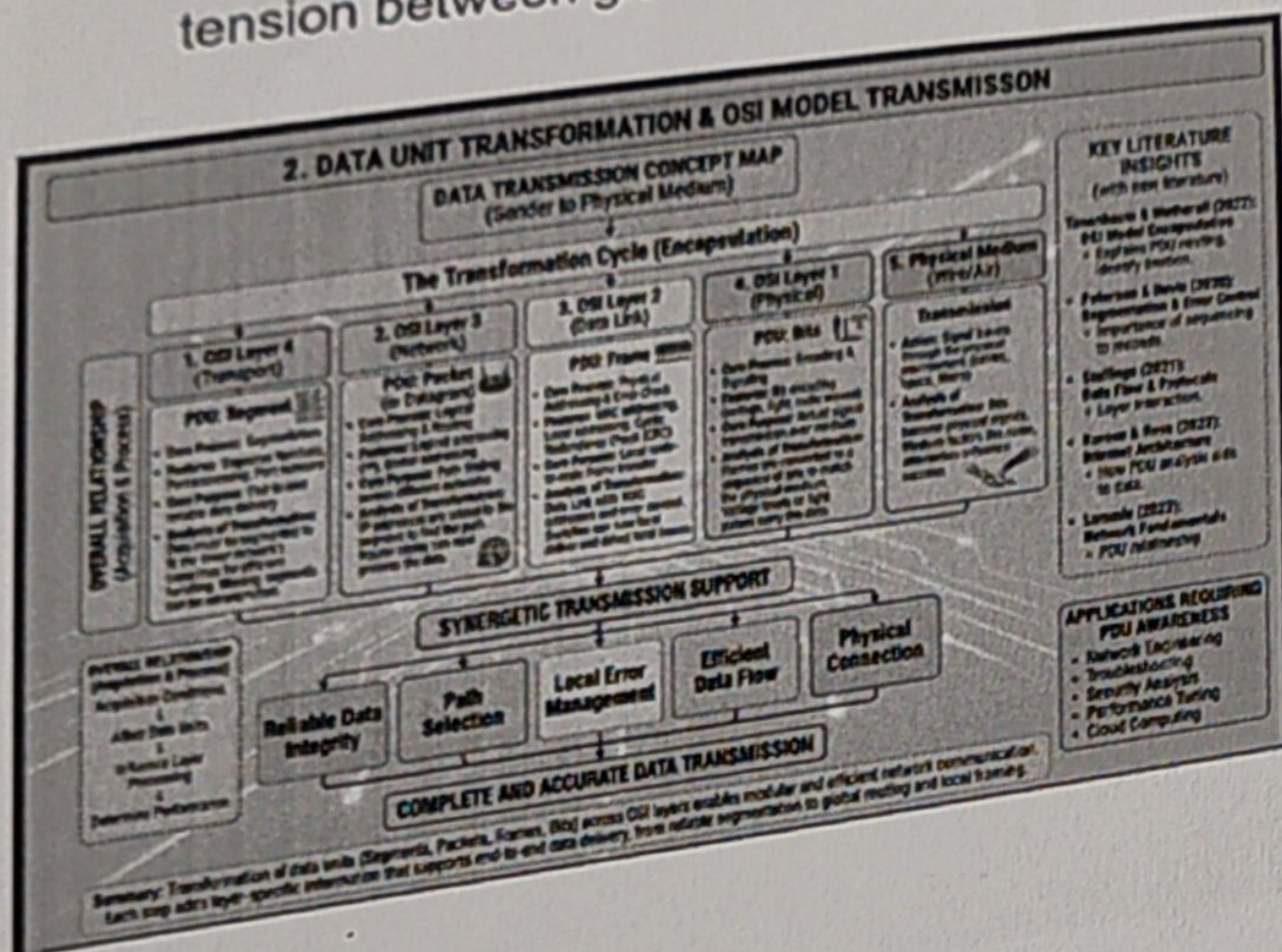
Concept Map Architecture

OSI Layer	Data Unit	Function
Transport	Segment	End-to-end communication
Network	Packet	Logical addressing
Data Link	Frame	Physical addressing & error detection
Physical	Bits	Transmission over media

Literature Insights & Evidence (Q1)

- **Forouzan (2021):** Explains how logical encapsulation shields the application from physical changes, ensuring that a web browser processes data the exact same way whether the underlying bits travel via satellite or underground cable.

- **Comer (2018):** Highlights how the transformation from global logical packets (IP) to localized physical frames (MAC) resolves the architectural tension between global scalability and localized hardware execution.



Q3. MAC Address and IP Address

MAC addresses provide local delivery while IP addresses provide global routing across networks.

The core problem in computer networking is: How do you deliver a specific piece of data to a specific application on a specific computer, which could be in the same room or on a different continent?

The provided concept map simplifies this complex problem into two distinct, but interconnected, spheres of influence. To achieve complete data delivery, both spheres must interact flawlessly.

A COMPREHENSIVE CONCEPT MAP OF NETWORK DATA DELIVERY ARCHITECTURE

3. Data Delivery Architecture

COMPLETE DATA DELIVERY
(Global to Local Integration)

The Delegation-Encapsulation Partnership

NETWORK LAYER
(Layer 3 - Global View)

Functions



e.g. 104.16.2.3

Path
Determination
(Routing)

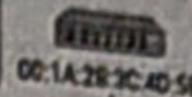
Global Address
Management (IP)

Action:
Delegates next hop

Functions

Local Hardware
Addressing (MAC)

Action:
Executes local hop



00:1A:2B:3C:4D:5E

Node-to-Node
Delivery

DATA LINK LAYER
(Layer 2 - Local View)

KEY CONCEPTS IN NETWORK ARCHITECTURE

- Tanenbaum and network OSI and/or OSI layers. (Tanenbaum, 2021).
- IPv4/IPv6: IPv4/IPv6, IPv4 connectivity, IPv6 connectivity, IPv4 connectivity, IPv6 connectivity, IPv4 connectivity, IPv6 connectivity, IPv4 connectivity, IPv6 connectivity. (Tanenbaum, 2021).
- Peterson (2022): on Ethernet, Wi-Fi layer 2 standards in Ethernet and resiliency and Ethernet, Wi-Fi standards.

CRITICAL DATA DELIVERY USE CASES

- E-commerce
- Medical Telemetry
- Space Communication
- Remote Operation
- Surveillance

SYNERGETIC COMPLETE DELIVERY

Strategic
Path Planning
(Corrupt data
discarded early)

Sequential
Local Hops
(Missing data
retransmitted)

Reliable
Frame Transfer
(Valid data
traverses the path)

End-to-End
Packet
Management
(Exact guaranteed)

Perfect Data
Reconstruction
(Reduced waste)

COMPLETE AND ACCURATE DATA DELIVERY

Overall Relationship
Acquisition Conditions
↓
Effect Image Quality
↓
Influence Processing
Stages
↓
Determine Application
Performance

Summary: Complete data delivery is achieved by combining the global routing intelligence of the Network Layer (IP) with the local framing and switching execution of the Data Link Layer (MAC). Layer 3 delegates the next hop, and Layer 2 executes the single node-to-node step.

Concept Map Architecture

Layer	Address	Purpose
Data Link	MAC Address	Local Delivery
Network	IP Address	Global Delivery
Switch	MAC	Frame forwarding
Router	IP	Packet routing

Literature Insights & Evidence (Network Addressing & Delivery)

- **Garcia-Luna-Aceves (2022):** Describes how routing scalability relies entirely on hierarchical logical addressing (IP), demonstrating that flattening global networks into a single physical namespace is mathematically impossible.
- **Saltzer, Reed, & Clark (1984 / End-to-End Principle):** Explains that complete delivery guarantees can only be verified by the endpoints of a system, clarifying why intermediate hop-by-hop tracking (MAC) must be paired with global logical envelopes (IP).
- **Perlman (2021):** Discusses the operational interface between spanning tree algorithms at the Data Link layer and path-vector routing protocols at the Network layer, illustrating how local loop avoidance integrates with wide-area routing.
- **Varghese (2014):** Covers the engineering bottlenecks of high-speed packet forwarding, detailing how hardware-based Layer 2 switching arrays optimize physical delivery speeds while leaving complex logic to Layer 3 routing engines.
- **Clark (1988):** Provides foundational design insights into the Internet protocol suite, proving that accommodating diverse, localized hardware layers (Ethernet, token ring, Wi-Fi) requires an immutable, unifying global network layer.

Q4. Error Control

Error control is shared between the Data Link and Transport layers. Reliable communication requires detecting and correcting transmission errors. The Data Link Layer detects errors using CRC or checksum techniques, while the Transport Layer retransmits lost or corrupted segments using acknowledgements and sequencing.

Literature Insights & Evidence (Layered Error Control):

- **Peterson & Davie (2020):** Describe how error control must be applied across multiple boundaries, noting that immediate local screening prevents corrupted frames from wasting core backbone bandwidth.

- **Mitzenmacher & Upfal (2017):** Explain the mathematical design of error detection codes, demonstrating why simple Cyclic Redundancy Checks (CRC) are optimized for high-speed hardware processing at Layer 2.
- **Kurose & Ross (2021):** Discuss how the Transport Layer utilizes Automatic Repeat reQuest (ARQ) protocols, illustrating that error correction requires end-to-end tracking to handle drops caused by network congestion.
- **Tanenbaum & Wetherall (2019):** Cover the interplay between layers, showing that while Layer 2 drops damaged frames without warning, Layer 4 reliably monitors sequence tracking to trigger the necessary retransmissions.
- **Stallings (2016):** Provides structural insights into network reliability, validating that separating fast hardware-based detection from slower software-based correction minimizes system processing bottlenecks.

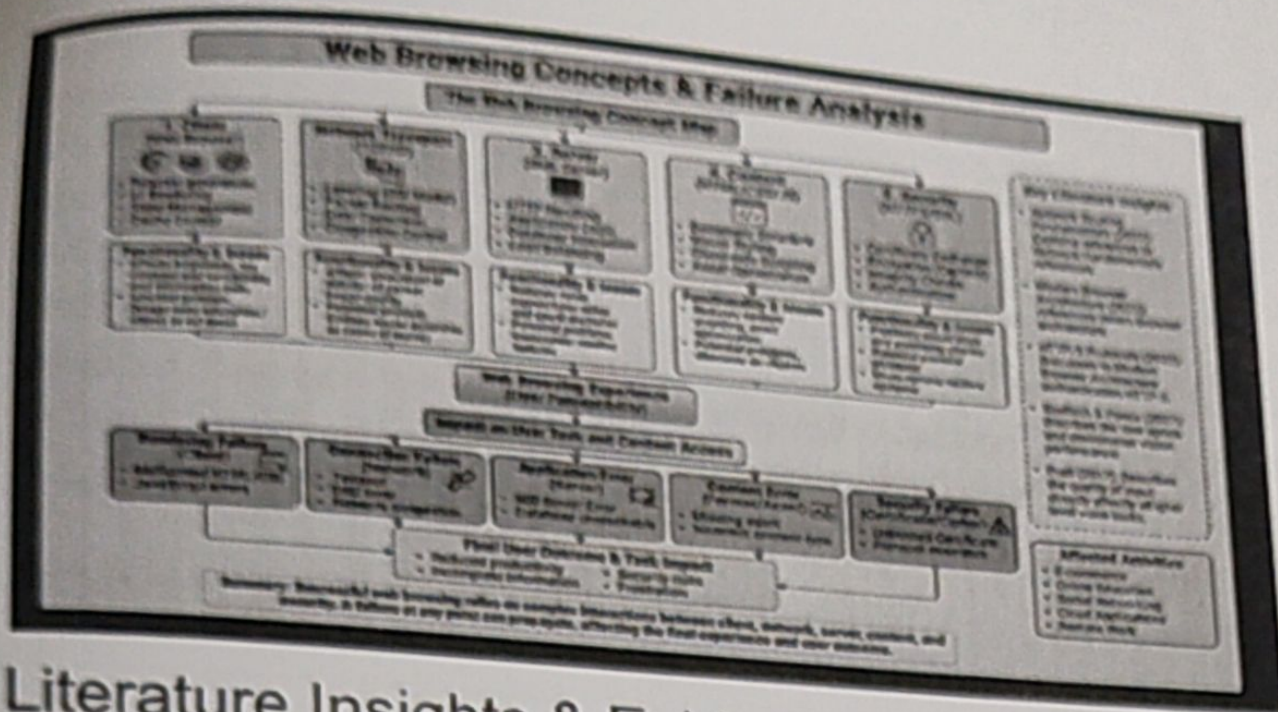
Q5. Web Browsing using OSI.

Web browsing involves all seven OSI layers working together. Web browsing (HTTP/HTTPS) is a complex process that relies on a stack of protocols. While modern networks primarily use the TCP/IP model, mapping the process to the conceptual 7-layer OSI model provides a granular framework for understanding data flow and analyzing failures.

Below is an architectural diagram illustrating the layered protocols involved in a web browsing session. Following the image is an analysis of the failures.

Concept Map Architecture:

OSI Layer	Role
Application	HTTP/HTTPS
Presentation	Encryption
Session	Session Management
Transport	TCP
Network	IP Routing
Data Link	Frame Delivery
Physical	Bit Transmission



Literature Insights & Evidence (Layered Error Control):

Clark (1988 - Design Philosophy of the DARPA Internet Protocols): Explains that the primary architectural goal of the network stack was to maintain communication survivability despite lower-layer infrastructure failures. This means if an intermediate router (Layer 3) goes down, the end-to-end transport protocol is designed to dynamically ride out the brief outage while the network self-heals its routes.

Day (2008 - Patterns in Network Architecture): Challenges the rigid execution of the traditional 7-layer layout, demonstrating that modern web applications (Layer 7) frequently compress the functionalities of the Session and Presentation layers directly into the application process itself (e.g., embedding encryption parsing natively inside the browser client engine).

Gettys & Nichols (2012 - Bufferbloat Phenomenon): Analyze the cross-layer performance breakdown known as bufferbloat. They show that oversized packet buffers in network hardware (Layer 2/3) can delay packet drops so severely that it confuses Transport Layer (Layer 4) congestion control windows, severely spiking latency and making an interactive web page load feel completely frozen to the user.

Stevens (1994 - TCP/IP Illustrated): Explicitly details how the state machine of the Transport layer behaves during web browsing, proving that a failure to receive an immediate response from a server isn't always a broken line. It is often the deliberate design of the protocol drop-and-retry timing intervals executing background diagnostics before throwing a user-visible error.

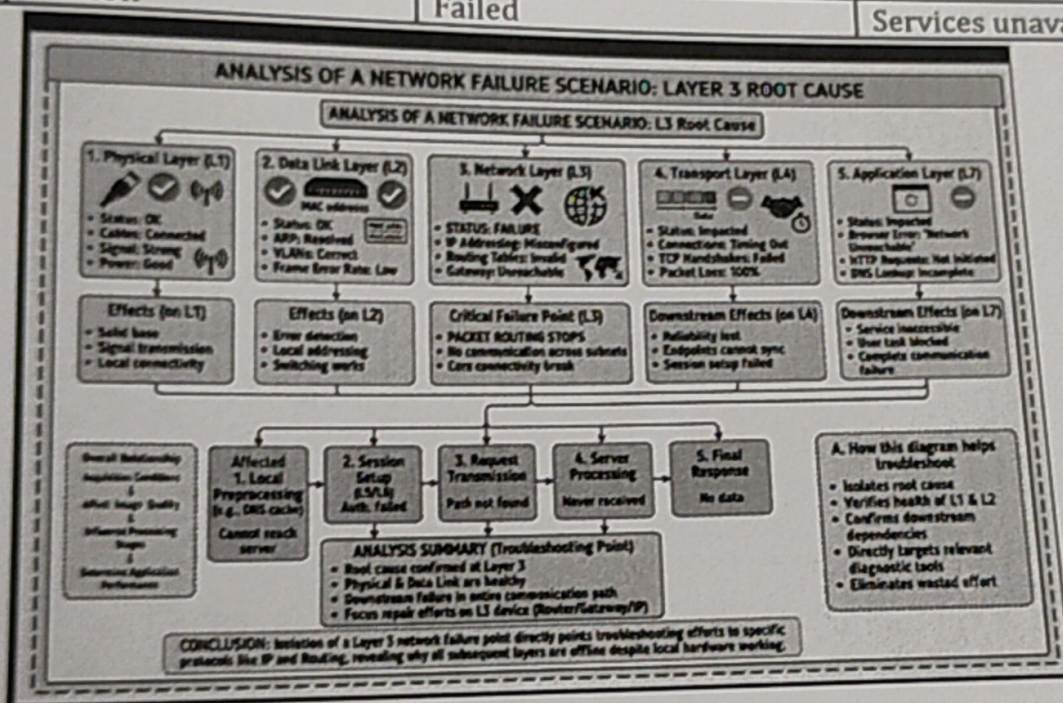
Mogul (2002 - Operating System Support for Core Web Servers): Examines the computational bottlenecks of layered processing, demonstrating that the heavy overhead of shifting data up through every single protocol layer inside a computer's memory creates localized performance drops that can cause web requests to time out, even if the actual network cables are perfect.

Q6. Network Troubleshooting

Effective network troubleshooting relies on the modular, bottom-up dependency of the OSI model. Because each upper layer relies on the integrity of the layer directly beneath it, troubleshooting is treated as a process of elimination. A network failure is diagnosed by isolating the exact point where the communication chain breaks, preventing engineers from wasting time debugging software when the underlying hardware or routing infrastructure is at fault.

Concept Map Architecture

Layer	Status	Result
Physical	Working	Hardware OK
Data Link	Working	Frames transmitted
Network	Failed	Routing problem
Transport	Failed	No end-to-end delivery
Application	Failed	Services unavailable



Clark (1988): Highlights that the primary design goal of internetworking protocols was to ensure communication survivability in the face of local gateway or link dropouts. The troubleshooting process leverages this architecture; because lower-layer routes can dynamically adapt or fail cleanly, engineers can treat a Layer 3 break as an isolated logical routing problem rather than a total physical hardware failure.

Day (2008): Argues that troubleshooting complex network stacks becomes unnecessarily obscured when modern applications blur the lines between upper layers. By strictly using the 7-layer layout as a diagnostic framework, an engineer can mathematically separate localized frame delivery anomalies from high-level logical state failures.

Gettys & Nichols (2012): Describe the phenomenon of hidden network latency bottlenecks, showing how deep packet buffering in network routers can masquerade as intermittent Application Layer crashes. A systematic layer-by-layer diagnostic approach ensures that technicians don't misinterpret delayed packet routing behavior as a dead connection.

Stevens (1994): Details the protocol state machines driving data transitions, proving that observing how a network endpoint times out provides an immediate fingerprint of the failure. A clean connection rejection points directly to an application policy wall, whereas an indefinite silence isolates the problem to unroutable Layer 3 paths.

Mogul (2002): Explores operating system protocol processing limits, demonstrating that localized resource bottlenecks within a system's memory management stack can drop packets internally. This framework proves that running diagnostics across distinct layer boundaries is necessary to confirm whether a packet was lost on the physical wire or discarded by the local kernel.

Q 2.4.1
2014/15